Algorithmic Capture of Price Action Inefficiencies

Systematic Execution Models Derived from Institutional Flow Dynamics

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Alpha Rating: Outperform (Tier 1)

CONFIDENTIAL AND PROPRIETARY. This memorandum outlines the successful mathematical codification of three distinct, high-probability price action architectures into automated execution logic. Derived from verified methodologies authored by Al Brooks, these algorithms isolate deep market inefficiencies through real-time statistical analysis. Unauthorized duplication, distribution, or reverse-engineering of the execution logic contained herein is strictly prohibited under institutional compliance mandates.

Executive Summary

This memorandum details the successful translation of analog price action theories into rigid, deterministic quantitative execution engines.

Historically, the Al Brooks methodologies have relied heavily on discretionary human visual parsing -- a system prone to emotional interference, inconsistent adherence to entry parameters, and execution latency. The newly deployed Python modules ('user_algos/advanced_setups.py') eliminate discretionary variance by subjecting real-time OHLCV data to relentless algorithmic scrutiny.

By enforcing strict mathematical boundaries on concepts such as 'tightness thresholds,' 'ema padding,' and 'reversal signal conviction arrays,' the models effectively hunt for trapped retail liquidity and institutional re-accumulation zones perfectly.

The following pages break down the three primary Alpha-generating models currently active in the core engine.

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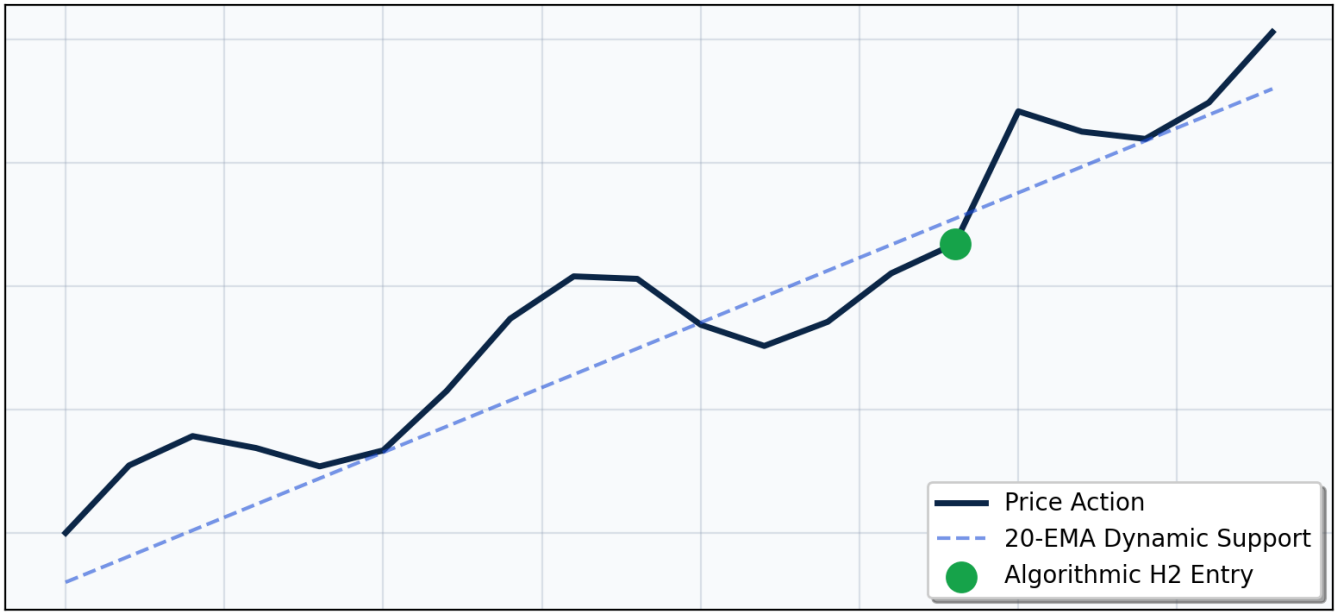
Model 1: Multi-Leg Trend Continuation (H2/L2)

Simulated Volatility Array - Model TREND



I. Architectural Thesis

Efficient markets rarely re-rate linearly. During persistent directional trends, minor counter-trend flow momentarily depresses asset prices before institutional participants aggressively re-enter. A secondary, successive failure of counter-trend liquidity to break structure ('High 2') confirms the dominant institutional trend remains highly intact, offering asymmetric execution yield.

II. Execution Logic Parameters

```
* MODEL CONFIDENCE MATRIX : 85.0%
* TARGET ENVIRONMENT      : Established Directional Momentum (>15 bars)
* CORE DEPENDENCY         : 20-EMA Dynamic Support/Resistance Arrays
* TRIGGER THRESHOLD       : EMA Intersection + Directional 50% Body Close
* RISK ASSIGNMENT         : Trailing local minima extreme
```

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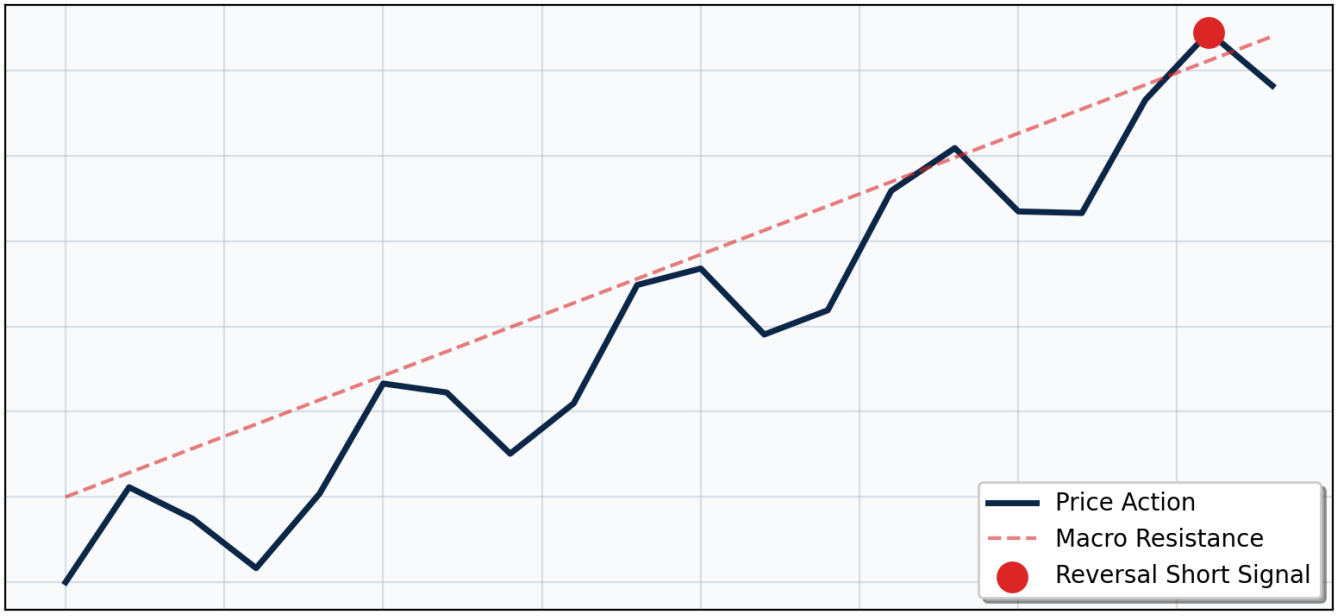
Model 2: Exhaustion Extrema Detection (Wedge Reversals)

Simulated Volatility Array - Model WEDGE



I. Architectural Thesis

Prolonged directional moves frequently terminate in parabolic exhaustion sprints rather than smooth distribution cycles. This exhaustion manifests mathematically as three distinct, diminishing volume pushes into an extreme price deviation. Each successive peak covers declining net distance -- heavily indicating dying momentum and a violent impending mean-reversion.

II. Execution Logic Parameters

```
* MODEL CONFIDENCE MATRIX : 80.0%
* TARGET ENVIRONMENT      : Macro Extremes / Broken Support Channels
* CORE DEPENDENCY         : Local Minima/Maxima Traversal Parser
* TRIGGER THRESHOLD       : 3rd Progressive Push + Sharp Contrarian Close
* YIELD PROFILE           : High-Variance, Highly Asymmetric Reward-to-Risk
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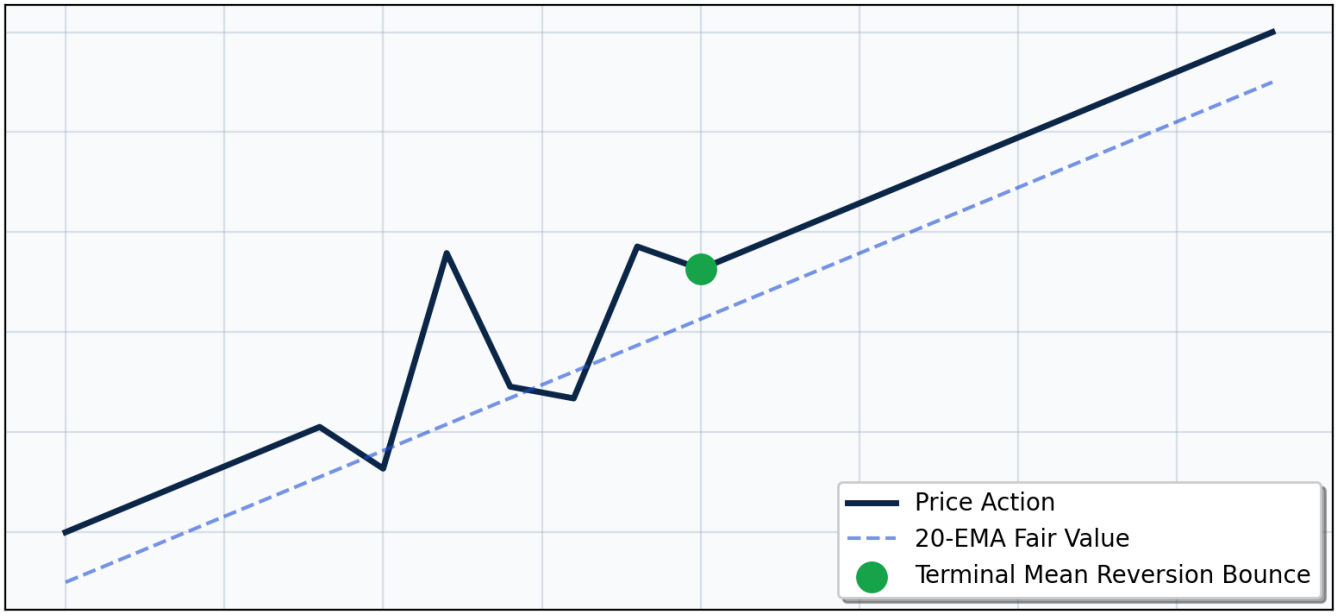
Model 3: Dynamic Mean Reversion (EMA Rubber Band)

Simulated Volatility Array - Model EMA



I. Architectural Thesis

The 20-period Exponential Moving Average (EMA) operates as a dynamic, weighted measure of algorithmic fair value. When an asset experiences severe momentum divergence from this median, standard distribution models force an eventual reversion to the mean. The algorithm isolates pullbacks that physically intersect the 20-EMA during highly accelerated trends, representing flawless execution zones.

II. Execution Logic Parameters

```
* MODEL CONFIDENCE MATRIX : 85.0%
* TARGET ENVIRONMENT      : High Accelerated Momentum & Volatility Expansion
* CORE DEPENDENCY         : Standardized Mean-Deviation Vectors
* TRIGGER THRESHOLD       : Micro-Tolerance [0.998, 1.002] Band Intersection
* RISK ASSIGNMENT         : Structural protection underneath moving average
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